



Checking a claim

Task A: Analysing headlines

1) A magazine article titled “Look at the savings you would make if tax was abolished” states the figures below.

Correct the prices of each item without VAT.

Television costing £450

Would cost only ~~£360~~ £375

(without 20% VAT)

÷ 1.2

%	Cost
120	£450
100	£375

Car seat costing £180

Would cost only ~~£171~~ £171.43

(without 5% VAT)

÷ 1.05

%	Cost
105	£180
100	£171.43

2) In a raffle there is a 1 in 2180 chance of winning a prize. A headline reads “Just buy 2 tickets and increase your chance of winning by 100%”.

a) What is the chance of winning with one ticket, as a percentage?

$$\frac{1}{2180} = 0.0004587... \quad 0.046\% \text{ to 2 significant figures.}$$

b) What is the chance of winning with two tickets, as a percentage?

$$\frac{2}{2180} = 0.0009174... \quad 0.092\% \text{ to 2 significant figures.}$$

c) What is the percentage point difference in the percentage chance of winning between buying 1 and 2 tickets?

Approximate percentage point increase of 0.046%

d) Explain why, although technically correct, the claim could be misleading.

It is mathematically accurate; doubling the number of tickets does double your chance of winning as 0.092 is double 0.046

This is the same as an increase of 100%.

The chance of winning is so small that doubling that chance is still only a very small percentage, the chance has only increased by 0.046 percentage points. People who do not understand probability may find this misleading.

e) Use trial and improvement to find how many tickets you would need to buy to increase the probability of winning to 1%.

$$\frac{21}{2180} = 0.009633... \quad \frac{22}{2180} = 0.01009... \quad \text{You would need to buy 22 tickets to increase the chance to 1\%.}$$

3) Explain the mistake in each headline.

- a) "Local employment rates fell by 10% last year and 15% the previous year. In just 2 years, employment rates have fallen by 25%."

When calculating two successive percentage changes you cannot just add the percentages together.

E.g. If there were 100 people employed one year then a decrease in 15% would be $100 \times 0.85 = 85$

Then decreasing this by 10% would give $85 \times 0.9 = 76.5$

This is a decrease of 23.5% not 25%

- b) "28% of men and 22% of women say they only clean their teeth once a day. A whopping 50% of adults are not cleaning their teeth twice a day."

This is not how percentages work. The percentages are of two different populations, if we assume the same amount of men and women the total percentage of people will be the mean average of the two (25%).

Also this was the number of people who said they brush their teeth once a day, there may be people who do not brush their teeth every day or brush their teeth more than twice. So the claim that 50% of people are not cleaning their teeth twice a day is definitely wrong.

- c) "Decrease your food expenditure by 300% a week by taking your own lunch to work (based on a takeaway lunch cost of £28 a week and home made lunch cost of £7 a week)."

A value cannot decrease by more than 100%.

It is true that increasing £7 by 300% is the same as multiplying by 4 which gives £28 but this does not work in reverse.

The calculation should be $7 \div 28 = 0.25$ which is a 75% decrease.

"It cost 4 times as much to buy a takeaway lunch" would be acceptable or "Decrease your food expenditure by 75%" (the 'each week' may also be misleading as the time scale doesn't affect the percentage).



Task B: Estimating to check

1) An article reads: “The average person will spend 100 000 hours on their phone in their lifetime.”

a) Estimate how long you think the average person will spend on their phone in a lifetime. Write down any estimations you make.

These are only approximate figures and the answer will depend largely on your estimation of how long most people spend on their phone a day.

If the average person spends 5 hours a day on their phone:

$$5 \times 365 = 1825 \text{ hours a year.}$$

$$\text{If average life expectancy is 80 years: } 1825 \times 80 = 146\,000 \text{ hours}$$

b) Does the article’s claim sound reasonable?

The claim is fairly reasonable, it’s potentially a little low considering the increase in use of technology such as mobile phones.

2) A newspaper states that “5% of the UK has a car parked on it.”

a) Estimate how many households there are in the UK.

Population of UK roughly 70 million and average of 2.4 people per household.

$$70\,000\,000 \div 2.4 = 29\,166\,667 \text{ so approximately 30 million households.}$$

b) Estimate how many cars there are in the UK.

If each household had 2 cars: 60 million cars in the UK

Note 2 cars per household is probably a high estimate but we haven’t included car impounds and showrooms etc.

c) Estimate how much space is taken up by these cars.

If a car measures 5 m by 1 m then it would take up 5 m²

$$5 \times 60\,000\,000 = 300\,000\,000 \text{ m}^2$$

d) Assume the area of the UK can be modelled by the two rectangles shown. What is the approximate area of the UK?

$$220 \times 1000 + 150 \times 100 = 235\,000 \text{ km}^2$$

e) Does the newspaper’s claim seem valid?

The tricky bit is changing m² to km²

$$1 \text{ km}^2 = 1000 \times 1000 = 1\,000\,000 \text{ m}^2$$

So cars take up 300 km²

$$\frac{300}{235000} = 0.00127... = 0.13\%$$

The article’s claim seems to be a massive over exaggeration.

